

Test-Time Training with Masked Autoencoders

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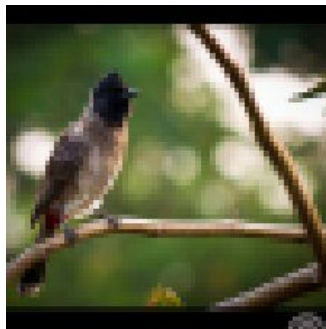
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A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Problem Introduction



Clean image

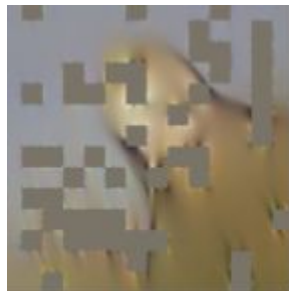


Corrupted images

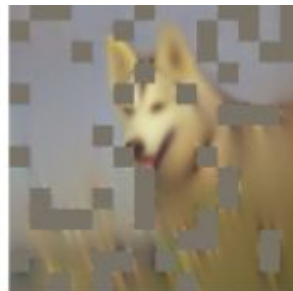
TTT MAE Approach



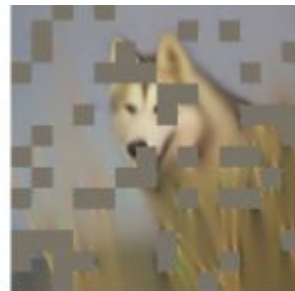
Original Image



Step 0



Step 50



Step 500

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Reproduction of the results

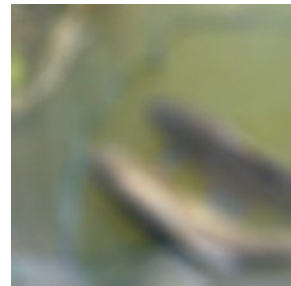
- 100 images per corruptions.
- Corruptions : defocus_blur, pixelate, brightness and gaussian_noise.
- 3 levels of corruption : 3, 4, 5.
- 20 steps per test image.



level 3



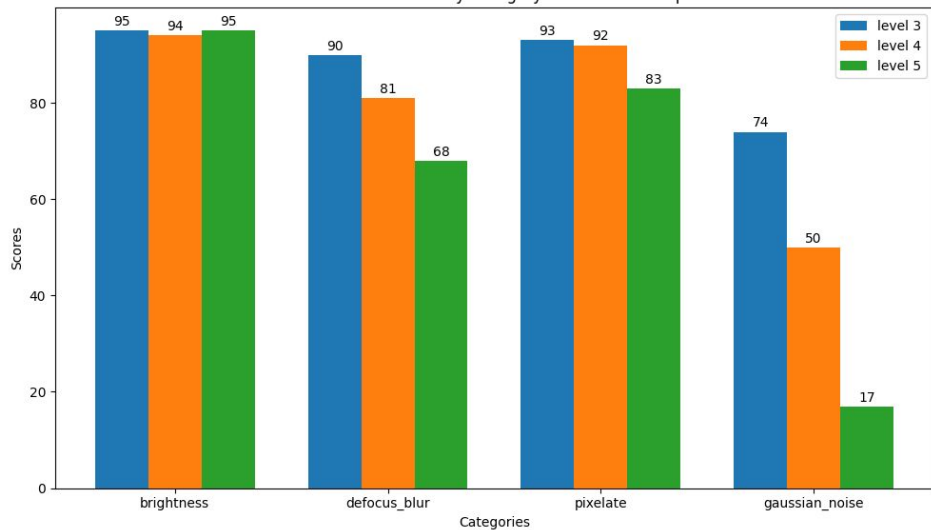
level 4



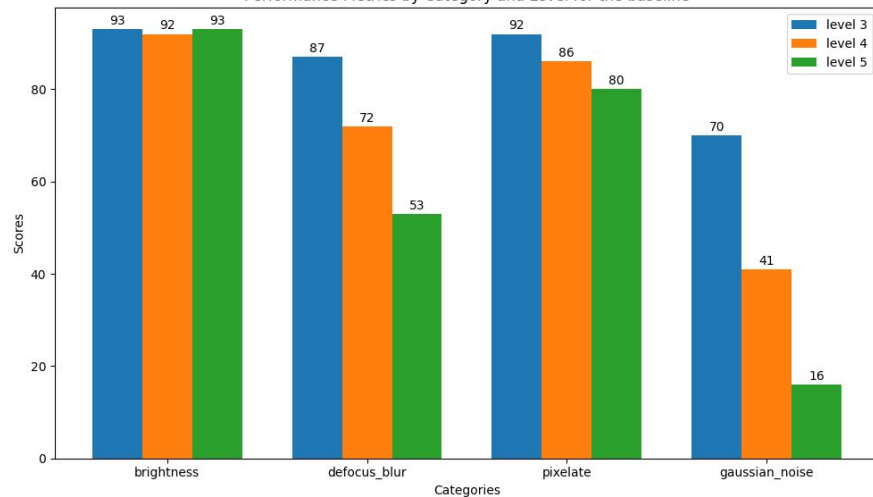
level 5

Reproduction of the results

Performance Metrics by Category and Level at Step 20



Performance Metrics by Category and Level for the baseline



Reproduction of the results

Accuracies (%) of the different models for level 5

	brightness	defocus_blur	pixelate	gaussian_noise
TTT-MAE (ours)	95	68	83	17
TTT-MAE	69.1	34.4	63	30.5
Baseline (ours)	95	53	80	16
Baseline	68.3	24.2	49.7	17.4

Impact of the steps

Accuracies (%) of TTT-MAE per steps for level 5

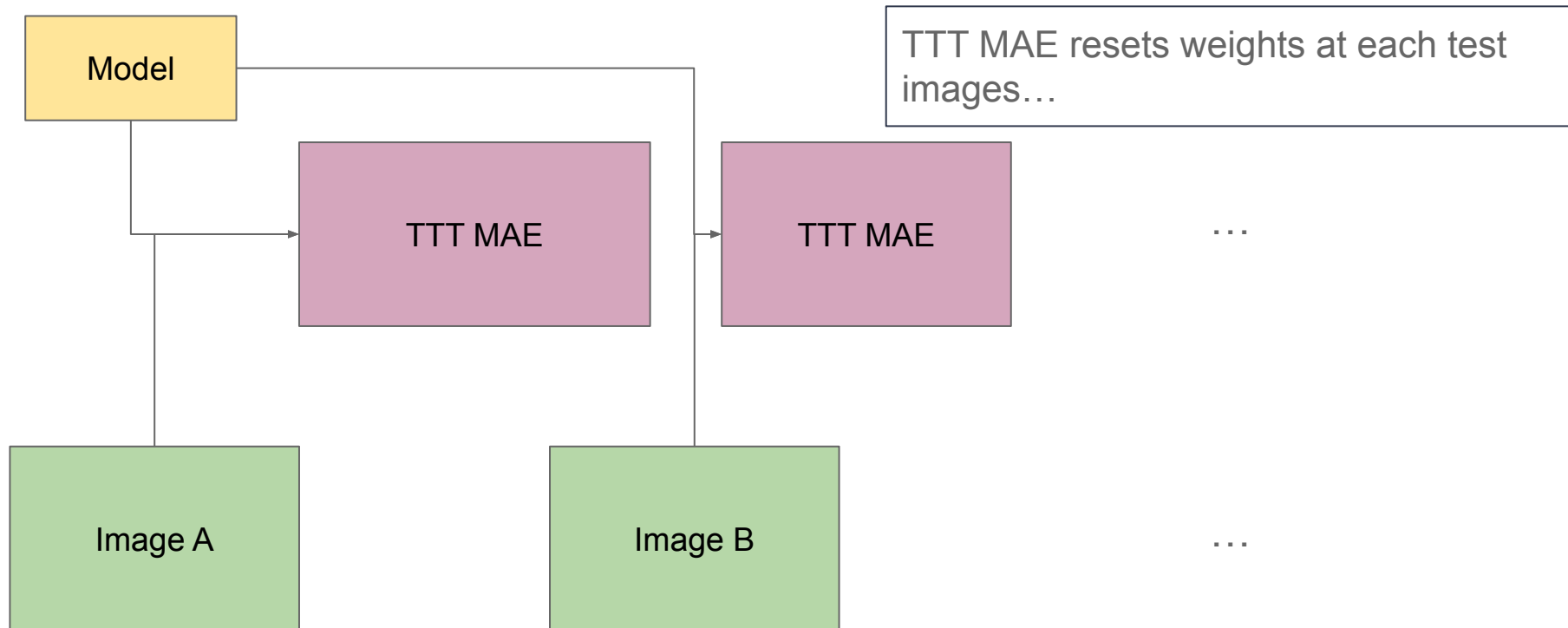
	brightness	defocus_blur	pixelate	gaussian_noise
5	94	60	80	16
10	94	62	80	16
15	95	69	82	17
20	95	68	83	17

Impact of the steps

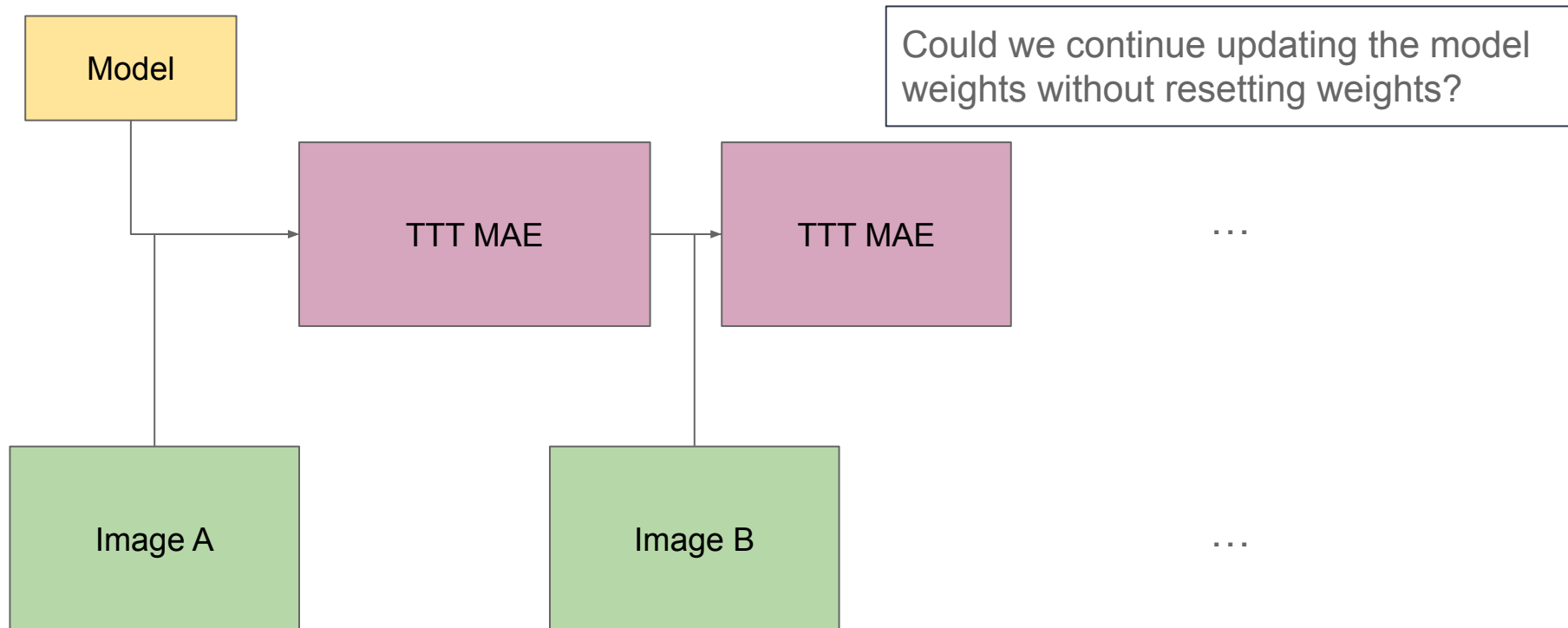
Accuracies (%) of TTT-MAE per steps for level 4

	brightness	defocus_blur	pixelate	gaussian_noise
5	92	72	86	41
10	92	76	88	45
15	93	78	90	48
20	94	81	92	50

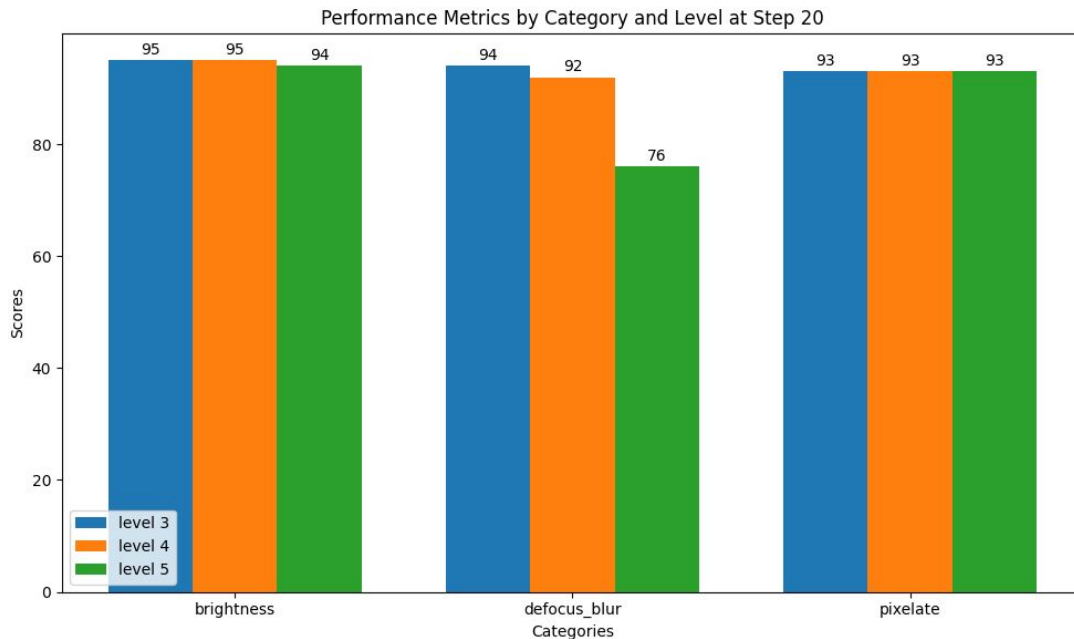
Online Approach



Online Approach



Online Approach



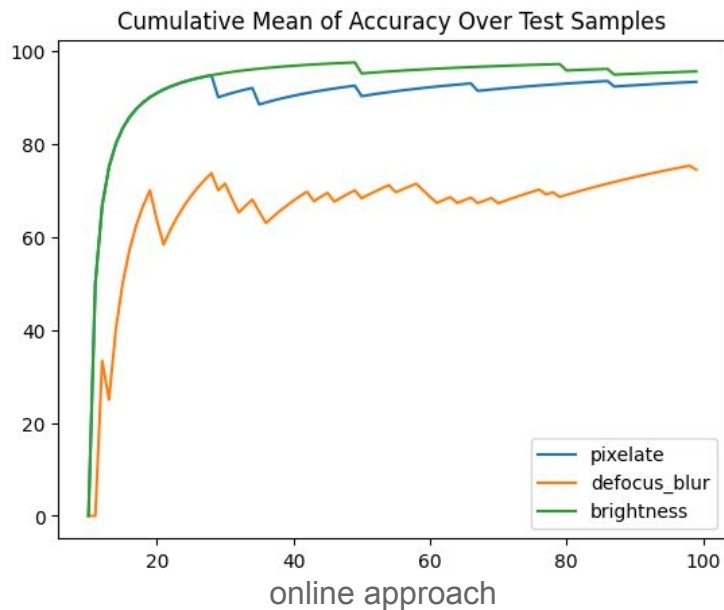
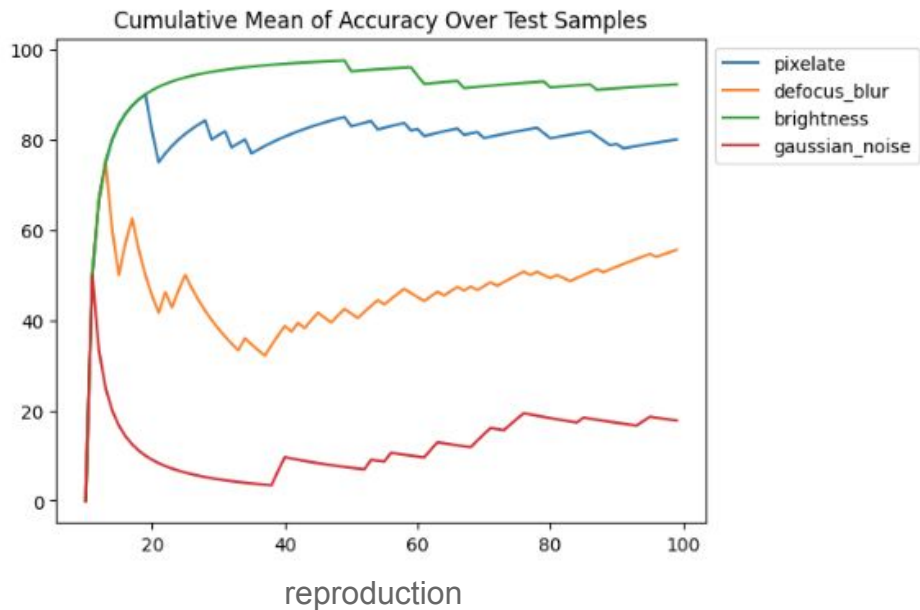
Online accuracy for 3 different categories and level

Impact of the steps

Accuracies (%) of online TTT-MAE per steps for level 5

	brightness	defocus_blur	pixelate
5	96	77	92
10	95	75	93
15	94	75	93
20	94	76	93

Online Approach



Online approach increases accuracy for new images without performing TTT MAE (step 0)

Adjusting the number of steps

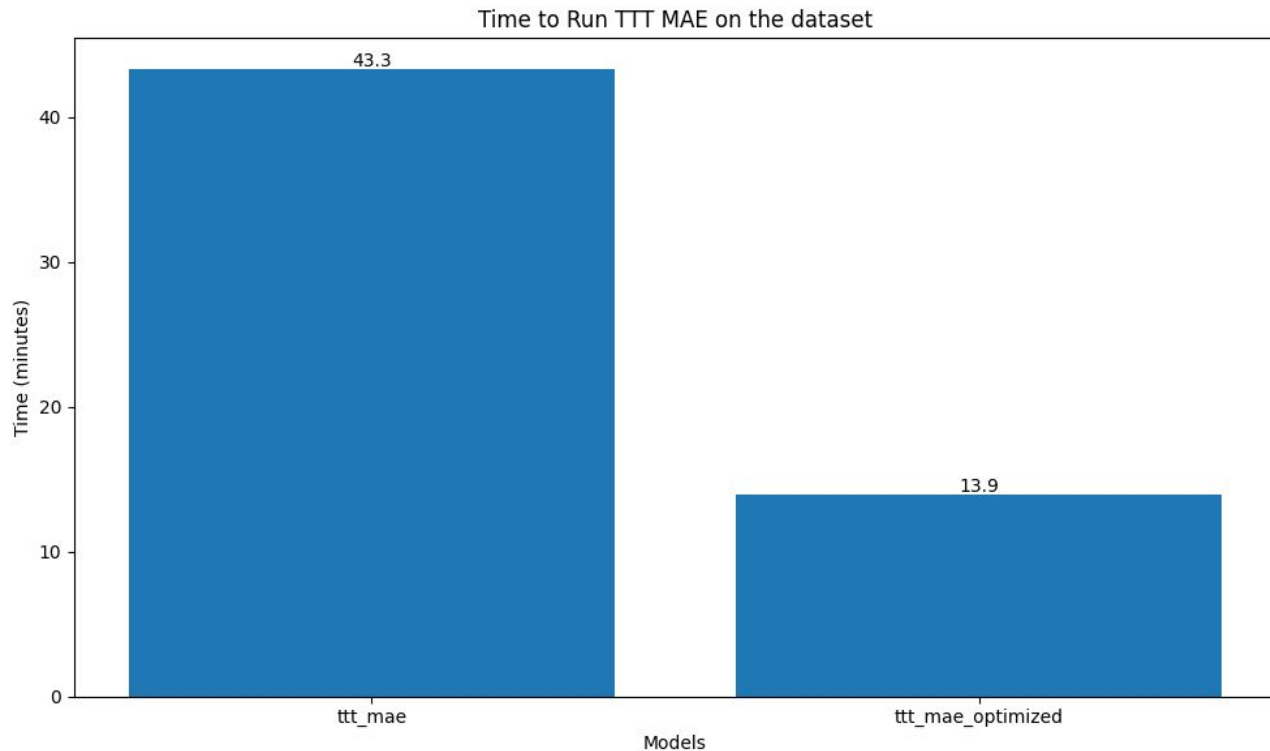
- TTT MAE has a fixed number of steps (20) for each test images.
- This is may not be efficient, if the model is working well for a certain corruption, we might not need to perform 20 steps.
- We propose to adjust the number of steps based on the confidence score for a test image.

Adjusting the number of steps

We will first predict on a test image and obtain the confidence maximum confidence score of the model. This score will be used to compute a new number of steps to perform for TTT MAE.

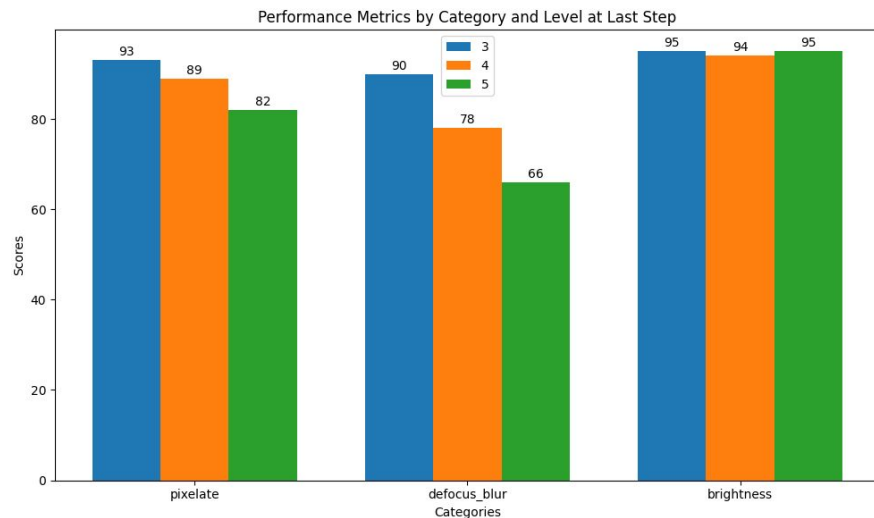
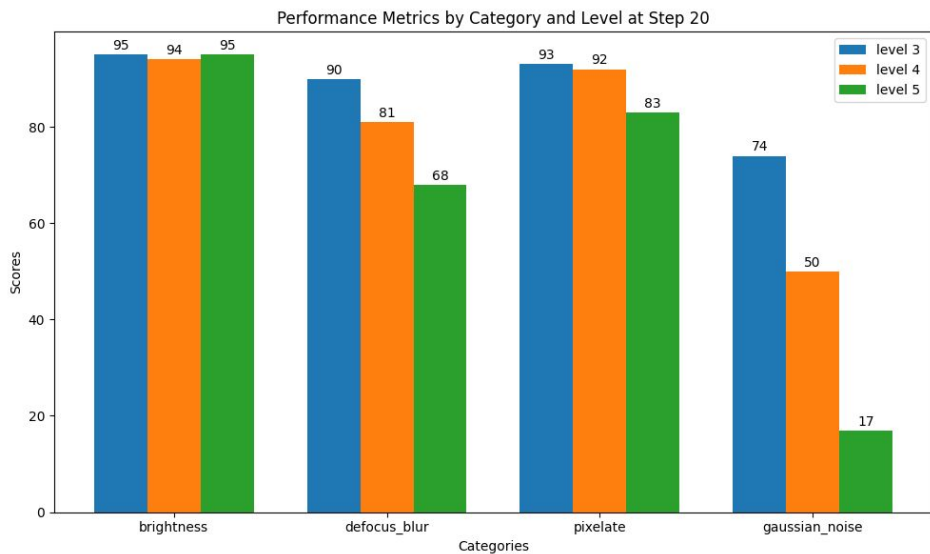
$$\text{maxSteps} = \max(1, \text{baseMaxSteps} * (1 - \max(\text{softmax}(\text{prediction}))))$$

Adjusting the number of steps



The optimized approach runs much faster than the base TTT MAE (3x faster)

Adjusting the number of steps



Our optimized approach is 3 times faster and does not impact much our metrics

Conclusion

Accuracies (%) of our experiments (mean over brightness, defocus and pixelated deformation)

	Baseline	TTT MAE*	Ours**	Online**	Optimized***
Level 3	90.66	69.4	92.66	94	92.66
Level 4	83.33	61.66	89	93.33	87
Level 5	75.33	53.1	82	93	81

* 10 steps all images

** 20 steps 100 images

*** Last ttt mae step 100 images

Issues

- Lack of compute
 - Less experiments
 - Much less images used therefore our results are a bit noisy
- TTT MAE framework did not allow to use different number of steps per images forcing us to rewrite the framework almost completely

Conclusion

- Reproduction of TTT MAE
 - Better/similar results
- Implementation of an online version of TTT MAE
 - Better first step performance over time
 - Better overall performance in the end for a given degradation
- Proposed a way to accelerate TTT MAE
 - 3x performance